Two-Dimensional Memtransistors: Modeling and Simulation for Circuit Design

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Introduction

- **Memtransistor Structure** : A three-terminal hybrid device (source, drain, gate) that merges memristor's non-volatile memory with transistor's gate-tunable conductance.
- **Independent Read/Write**: The gate terminal lets us control reading and writing separately, unlike memristors.
- **Enhanced Functionality** : Works as both memory and logic helps overcome the von Neumann bottleneck by reducing the gap between memory and processor.
- **Reduced Complexity**: Memtransistors eliminate the need for extra selector transistors that are required in memristor crossbar arrays to access individual memory cells. This leads to more area- and energy-efficient designs.

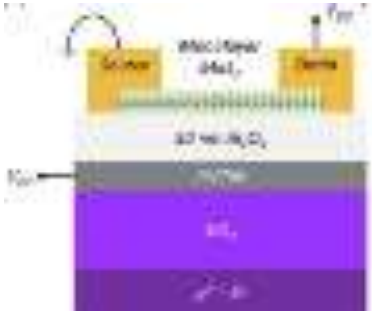

Figure: 2D Back gated memtransistor[1]

Need and Motivation

- **Need of Memtransistor**
  - **The von Neumann Bottleneck** : In traditional computers, CPU and memory are separate. Continuous data transfer between them slows performance and wastes energy.
  - **Energy Challenge :** For data-heavy tasks (e.g., AI, ML), this back-and-forth consumes enormous energy.
  - **Need for In-Memory Computing:** To solve this, computation must happen inside the memory itself, reducing data movement and improving efficiency.

- **Motivation**
  - By developing memtransistor-based systems, we can enable faster, smarter, and energy-efficient electronics for future AI and IoT applications.
  - High Scalability : The use of atomically thin MoS2 allows for aggressive dimensional scaling.
  - Brain-inspired neuromorphic computing for applications like real-time seizure detection.
  - Chip Fingerprints (PUFs): Tiny, unavoidable variations during the manufacturing of MoS2 create a unique physical ”fingerprint” for every single chip. This identity is impossible to clone, preventing counterfeit copies.

Literature Review

Working Mechanism of Memtransistors

- Operate mainly through charge trapping at MoS2/oxide interface and in oxide near interface (storing charges in traps) or resistive switching (changing resistance states).
- These mechanisms allow them to store information non-volatilely.

Role of 2D Materials

- Molybdenum Disulfide (MoS2) is one of the most promising channel materials for memtransistors.
- Its atomically thin structure makes it suitable for ultra-scaled devices.
- MoS2 is only 0.7 nm thick (monolayer).
- Enables smaller and denser devices compared to silicon.

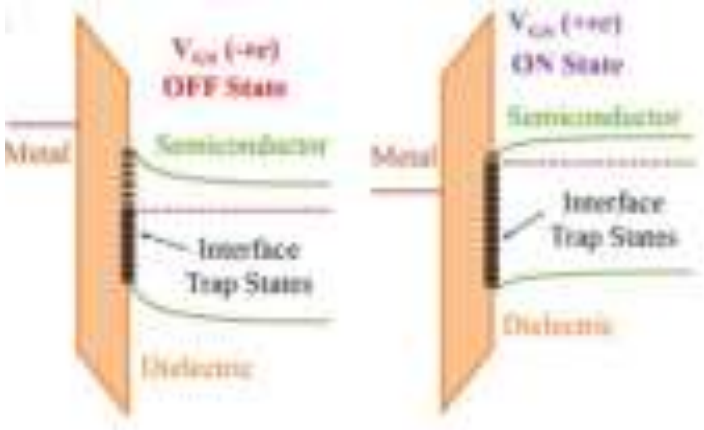

Figure: Trapping and Detrapping[3]

Cont.

Hardware implementation of Bayesian network based on two-dimensional memtransistors
(Publication: Nature Communications (2022)

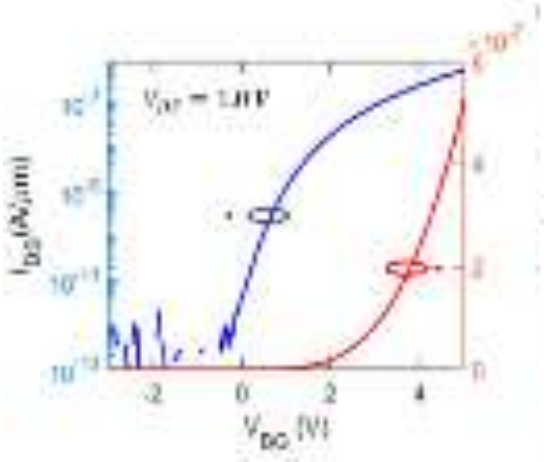

Figure: Transfer characteristics, i.e., source-to-drain current (IDS) vs(VBG), (VDS) of 1 V, in linear and logarithmic scale[1]

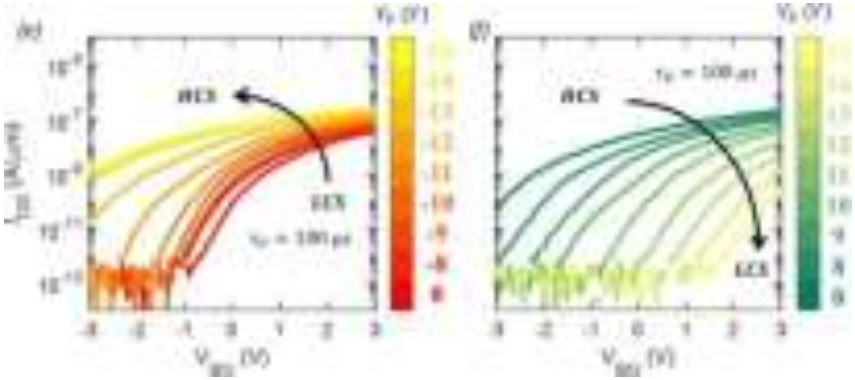

Figure: Id vs Vgs post programming pulse .Experimentalresult Log scale(X axis: gate voltage, Y axis Drain current)[1]

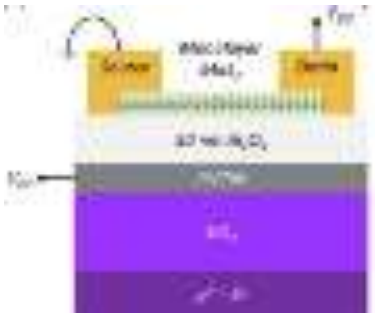

Figure: 2D-MoS2 based Memtransistor[1]

- Channel: Monolayer Molybdenum Disulfide (MoS2) .
- Gate: A local back-gate structure with a 50 nm Aluminum Oxide (Al2O3) charge trapping dielectric on a 40/30 nm Platinum/Titanium Nitride (Pt/TiN) electrode.
- Contacts: Nickel (Ni) and Gold (Au) are used for the source and drain.
- Substrate: The device is built on a SiO2/p++-Si substrate.

Dielectric Engineering: HfO_2 vs. Al_2O_3

- **Stronger Gate Control (High-k Effect):** HfO_2 possesses higher dielectric constant (ϵ_r) this significantly increases the gate capacitance, allowing the stronger gate control over the remote MoS_2 channel.
- **Intrinsic Defectivity:** The crystal structure of HfO_2 is inherently unstable, making it highly prone to forming stable oxygen vacancies during deposition.
- **Enhanced memory window:** These structural vacancies in HfO_2 function as deep **bulk** charge traps, enabling a far higher volume of storage and hence increase memory window.
- **Key Advantage of $Al2O_3$ over HfO_2 :** Due to $Al2O_3$ large bandgap (8.8eV), which minimizes leakage current and ensure stable conductance state.

Interface traps and bulk traps

- **Interface Traps state At MoS2/HfO2:** These traps sit directly at the MoS2–HfO2 interface and interact instantly with the channel through SRH capture/emission. They mainly degrade current and mobility but cannot store charge for long.
- **Oxide Traps state** :
  Located within the HfO2 dielectric volume, physically separated from the MoS2 bands. Can retain charge for long periods, making them the dominant mechanism enabling stable, non-volatile memory storage.

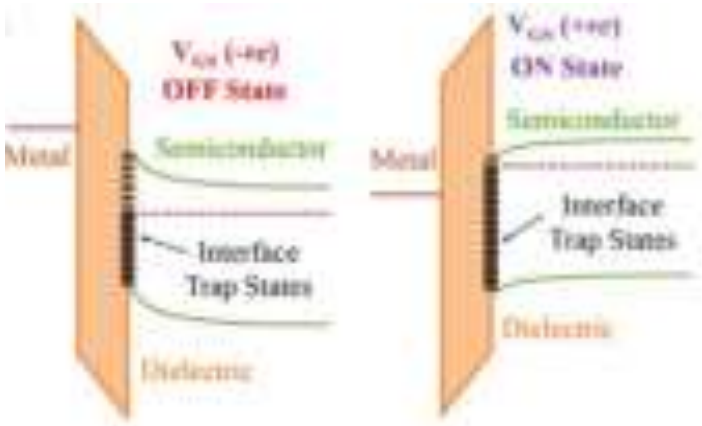

Figure: Interface trap states[2]

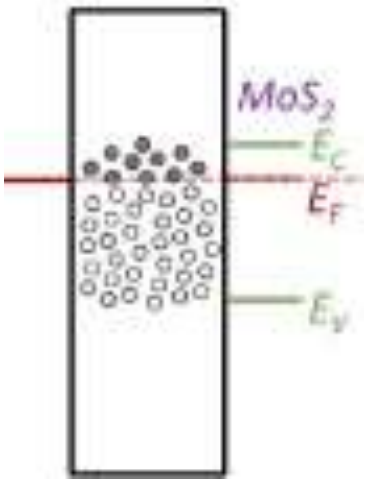

Figure: Oxide trap states[1]

Research Gaps

Experimental-Based Challenges

- It's still hard to make large, uniform, defect-free wafers of 2D materials like MoS2
- Fabrication imperfections cause variability in performance(Device to Device).
- Forming low-resistance electrical contacts with atomically thin channels is difficult.High contact resistance wastes power and reduces speed,performance.

Simulation-Based Research Gaps

- Need precise modeling of material properties and interface physics before fabrication .
- Need models that capture charge traps, contact resistance, and defects to bridge the gap between ideal simulations and experiments.

Current work

Device structure :

- The device is a three-terminal transistor-like structure based on monolayer MoS2..
- Substrate (SiO2 / p++Si) 285 nm SiO2 grown on heavily doped p++ silicon.
- Local Back-Gate (Pt 40nm TiN 3Onm with HfO2 25nm): The back-gate terminal is where programming and erasing pulses are applied.
- Channel (Monolayer MoS2 0.65nm): 2D material thickness (sub-nm scale) ensures excellent gate electrostatics.
- A 40nm Titanium metal is used as source and drain .

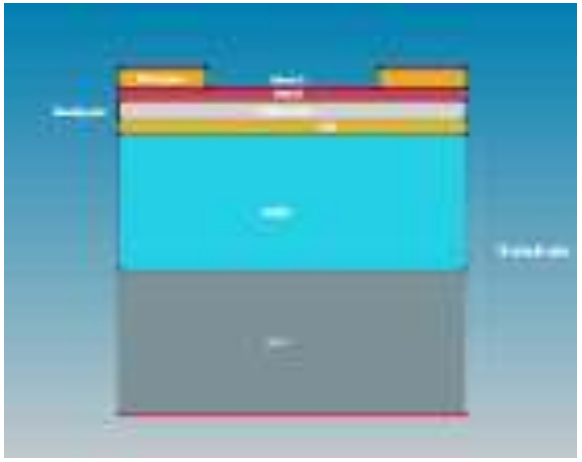

Figure: 2D Device structure sde

TCAD Physics models used

Trap Location	TCAD Model	Role in Operation	Trapping Physics	Parameters Used
MoS_2/HfO_2 Interface	Donor uniform, Acceptor uniform (10^{12} cm^{-2})	Primary function is to cause mobility degradation.	Shockley–Read–Hall (SRH) kinetics (local capture)	1.0×10^{-15} cm^2 cross-sections
HfO_2 Bulk (25 nm)	Acceptor Gaussian (e^- trap)	Erase operation; stores negative charge (5×10^{18} cm^{-3})	eBarrierTunneling Poole–Frenkel (Detrapping)	E_{mid} = 1.0 eV from E_C
HfO_2 Bulk (25 nm)	Donor Gaussian (h^+ trap)	Program operation; stores positive charge (1×10^{18} cm^{-3})	hBarrierTunneling and Poole–Frenkel (Detrapping)	E_{mid} = 1.5 eV from E_V

Table: Summary of trap locations, TCAD models, trapping physics, and parameters used.

Simulation result

Id vs Vgs characteristic

Parameter	Al_2O_3 Device	HfO_2 Device
Permittivity (ε_r)	$\approx$ 9	$\approx$ 25
Threshold Voltage (V_{th})	$\approx$ 1.7 V	Lower 1.2V
I_{ON}	$\approx 6 \times 10^{-7}$ A/μm	1×10^{-5} A/μm
I_{OFF}	$\approx 10^{-13}$ A/μm	slightly higher 10^{-11} A/μm

Table: Comparison of Al_2O_3 and HfO_2 Devices

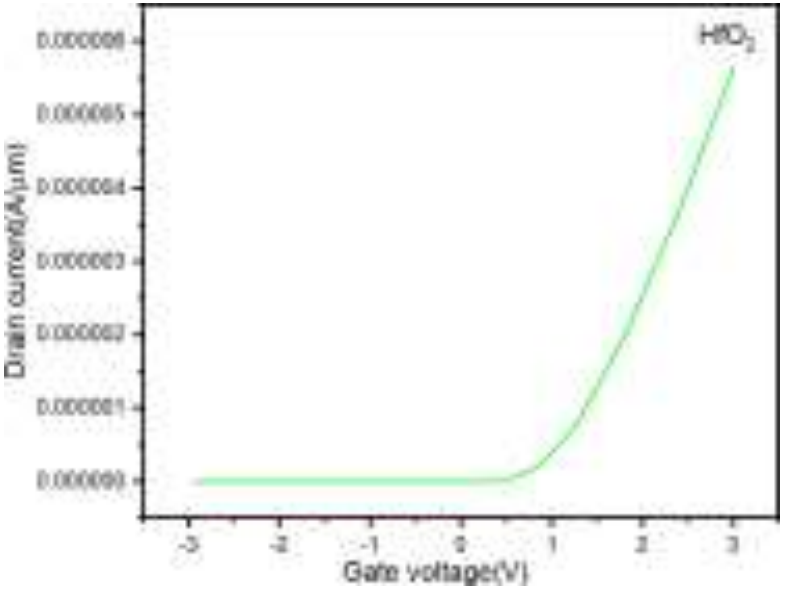

Figure: Ids vs Vgs linear scale

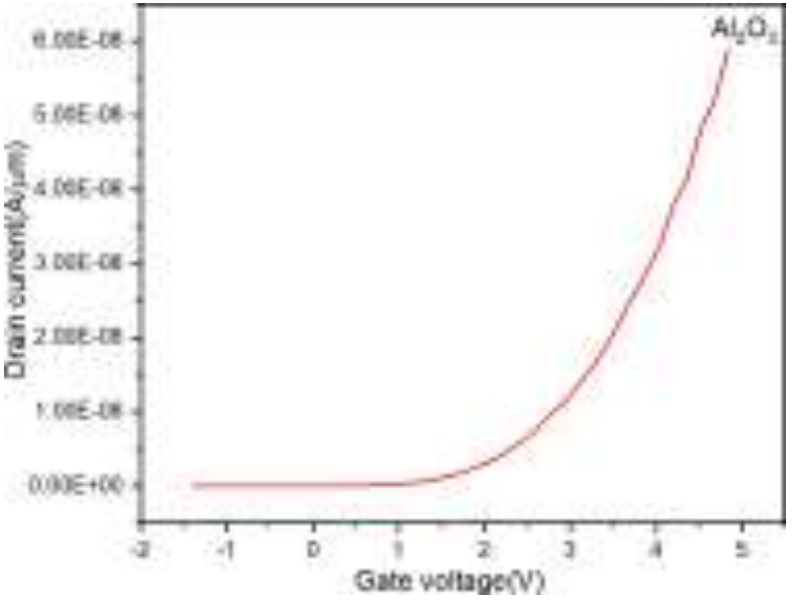

Figure: Id vs Vgs linear scale

Id vs Vgs characteristic at different trap concentration

Id vs Vgs characteristic

- Varied the trap concentration in the HfO2 bulk.
- Gate voltage sweep from -2v to 3v with Vds=0.05v..

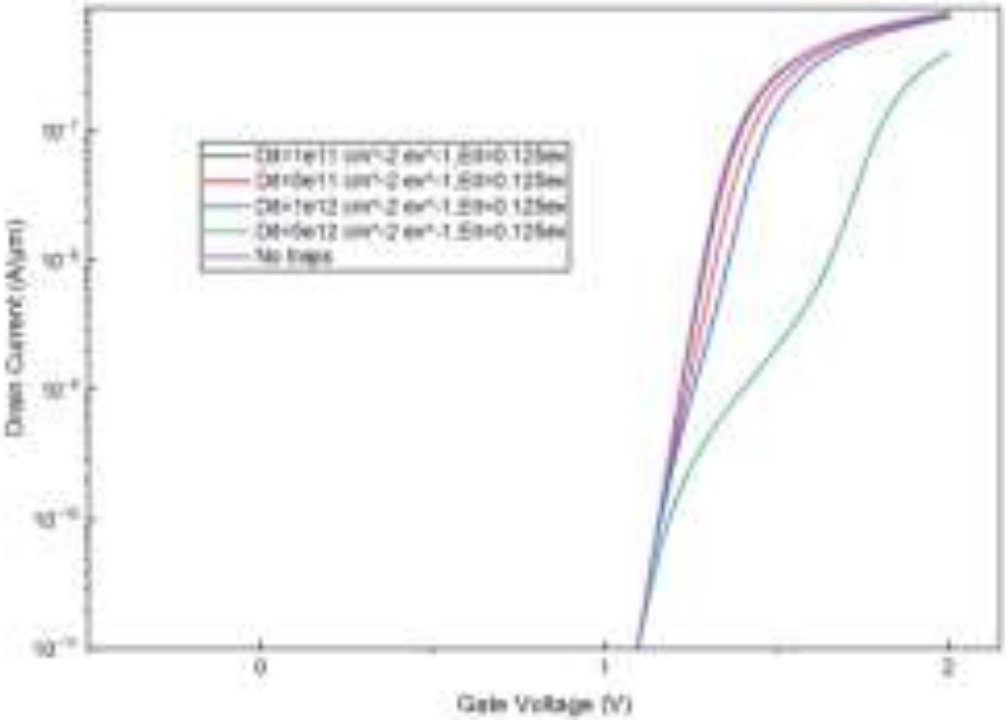

Figure: Drain current against TCAD simulation data at different values of *D*it

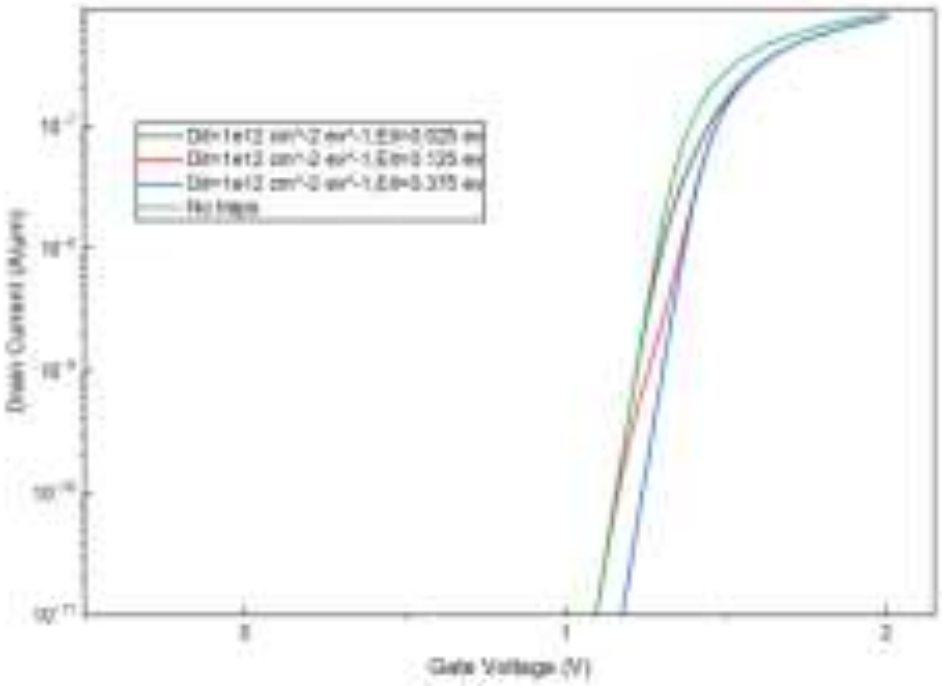

Figure: drain current against TCAD simulation data at different values of Eit

Post Programing pulse state

- VP voltage pulses of different amplitudes -7 V to -10 V applied to the local back-gate electrode, each for a duration of P=E = 100 µs,
- Post progammed state Id vs Vgs . .

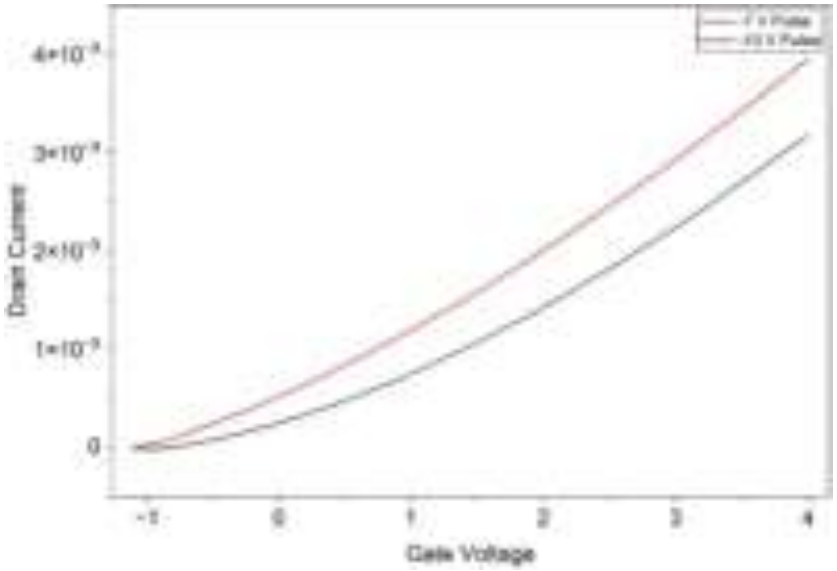

Figure: Ids vs Vgs linear scale post program state

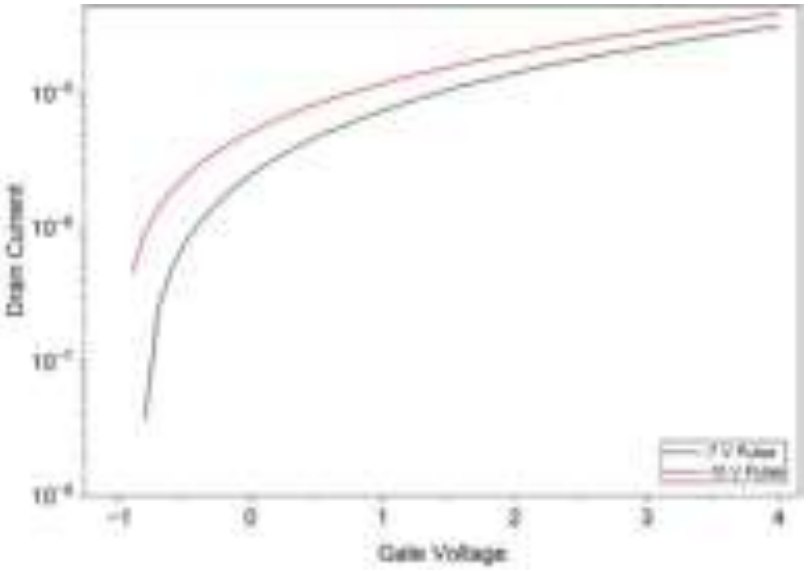

Figure: Id vs Vgs log scale post program state

Challenges in Current work

The following are the current challenges:.

- It's very difficult for the simulation to accurately capture the effect of very fast and powerful voltage pulses (programing and erase).
- Creating a mesh that is fine enough for numerical stability without making computational cost .
- Non linear physics causes the solution to diverge rapidly.

References I

[1] Zheng, Y., Ravichandran, H., Schranghamer, T.F. et al. Hardware implementation of Bayesian network based on two-dimensional memtransistors. Nat Commun 13, 5578 (2022). https://doi.org/10.1038/s41467-022-33053-x

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[4] Boyan Jin, Zhenlong Wang, Tianyu Wang, Jialin Meng, Memristor-Based Artificial Neural Networks for Hardware Neuromorphic Computing, Research, 10.34133/research.0758, 8, (2025).